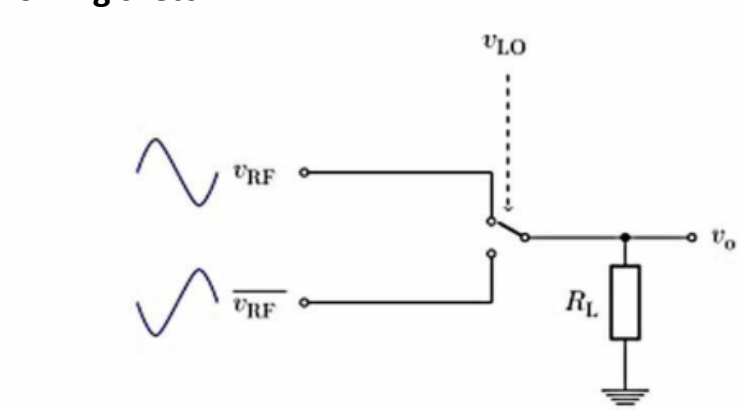
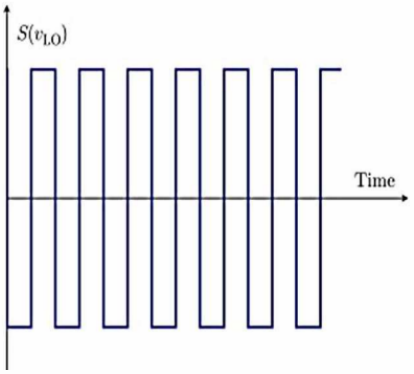
F3 Heterodyne Monitor
Double Balanced Mixer Concept

We can illustrate the concept of using a 4052 analogue multiplexer as a 'double balanced mixer' with the following sketch:



The opposite polarity ultrasonic signals VRF and $\overline{VRF}$ are the two inputs, and VO is the output.
The ultrasonic local oscillator output (VLO) is fed to the appropriate 4052 control input (pin 10).
In this case, alternate VLO polarity signals switch between the two RF signal inputs, and present these 2 signals alternately to the output.
The LO switching waveform can be represented by the square wave switching function: S(Vlo):



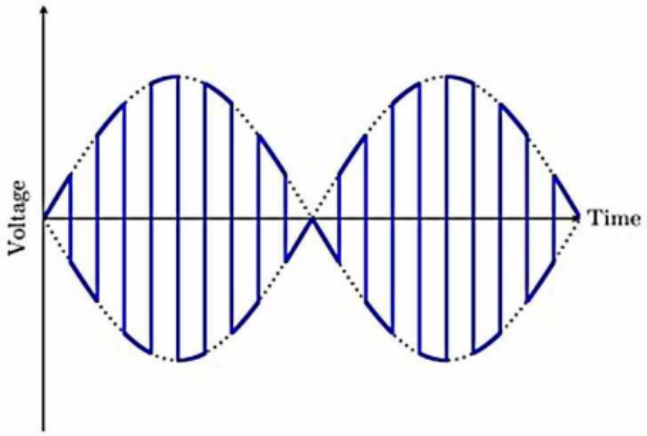
$$S(v_{LO}) = \begin{matrix} 1 \text{ if } v_{LO} \geq 0 \\ -1 \text{if } v_{LO} < 0 \end{matrix}$$

The Fourier series representation of the square wave switching waveform is given in the equation below, and consists of all the odd harmonics of the LO waveform, decreasing by 1/n.

$$S(v_{LO}) = \frac{4}{\pi}\sum_{n=1}^{\infty} \frac{\sin(n\pi/2)}{n} \cos(n\omega_{LO}t)$$

$$= \frac{4}{\pi}\cos(\omega_{LO}t) - \frac{4}{3\pi}\cos(3\omega_{LO}t) + \cdots$$

The resulting output voltage, Vo, is presented via R6 to the inverting input of IC3 (pin2) as the waveform shown below. This represents the product of VRF and S(Vlo) :



$$v_o = v_{RF} \cdot S(v_{LO})$$

$$= \frac{4}{\pi} v_{RF} \cos(\omega_{LO} t) - \frac{4}{3\pi} v_{RF} \cos(3\omega_{LO} t) + \cdots$$

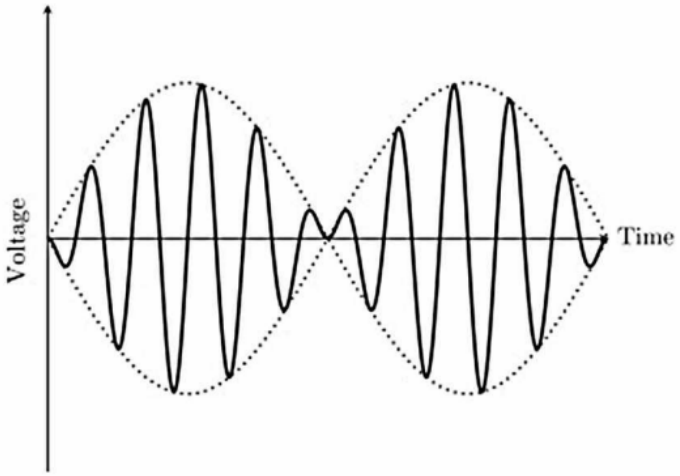
We set: $v_{RF} = V_{RF} \cos(\omega_{RF} t)$ **and consider only the 1st and 2nd terms, on the assumption that all higher order terms will be removed by the subsequent filtering.**
We can therefore write:

$$v_o = \frac{4}{\pi} V_{RF} \cos(\omega_{RF} t) \cos(\omega_{LO} t)$$

$$= \frac{2}{\pi} V_{RF} \cos[(\omega_{RF} + \omega_{LO})t] + \cos[(\omega_{RF} - \omega_{LO})t]$$

The output waveform therefore consists of two frequency translated versions of the RF signal, one centred at: $(\omega_{LO} - \omega_{RF})$ **and the other centred at:** $(\omega_{LO} + \omega_{RF})$**, both having an amplitude 2VRF/**π**.**

After low pass filtering the output waveform looks like this:



The waveform is typical of a double sideband suppressed carrier AM modulated signal.

The conversion gain of this double balanced mixer is 2/π **(c.-4dB), a c.6dB improvement over a single balanced mixer configuration.**

Roger Shore 3.26